

Moore's Law outlawed

Moore's law states that the amount of information storable on a given amount of silicon has roughly doubled every year since the microprocessor technology was invented. Andrew S. Grove, the Chairman of Intel, recently suggested that Moore's Law might be redundant by the end of the decade. The inactive Intel processors are leaking current, threatening the future valid-

ity of Moore's Law.

As chips become more powerful and draw more power, leakage tends to increase. Going by statistics, the industry is used to power leakage rates of up to 15 per cent, but chips constructed of increasing numbers of transistors can suffer power leakage of up to 40 per cent. Chips made up of a billion transistors may leak between 60 and 70



watts of power. The power is largely dissipated as heat, which causes cooling problems for powerful chips.

Currently, Intel is seeking ways to design chips with multiple cores with improved design and better insulators. The microprocessor giant intends to make more efficient use of the humble transistor in order to deliver ever increasing performance. In short, Moore's law is going to be upstaged by electricity!

Thought-provoking games

Researchers at the University of Boulder, located at Colorado in the USA, used video games to find out whether a cultural or racial stereotype could be linked to armed and unarmed males.

The subjects in the study were asked to shoot only when the human targets in the game

were armed; however, they typically shot at unarmed black men compared to unarmed white men. The video game showed photographs of males against a variety of backgrounds and asked the subjects to immediately conclude whether the image showed an armed or



unarmed man. The unarmed males had aluminium cans, cell phones, cameras and wallets. When asked to make a split-second decision, all the subjects concluded, mistakenly, that black men were armed.

The research helped establish that hidden cultural stereotypes exist in situations wherein the subject has to react very quickly and on instincts.

Over the hill

A 50-year-old computer called CSIRAC is still operational in Australia. This one-time state-of-the-art machine runs at a blistering 300 KHz, has 2 KB of RAM and 2.5 KB of storage. While the more rational among us may scoff at its utility, in its hey day this 7 tonne machine packed quite a wallop. Between 1951 and 1964, it worked with Australian universities on projects as diverse as calculating the design of water storage systems, weather prediction, Fourier analysis research and skyscraper design. CSIRAC also worked with Australian industry on studies for BP's Kwinana refinery, the Commonwealth Aeronautic Research Laboratory and was used by undergraduate and graduate students for research into programming.

snapshot

Online sales worldwide for Nov 1 to Dec 6, 2002, reached **\$8.2 billion**, up **29%** from a year

Source: ComScore

Drown in the music

Calling it the 'most definitive sound experience yet', Creative's Audigy 2 is a PC audiophile's delight. The Audigy 2's jaw-dropping specifications include 24 bit music fidelity with 96 KHz in seven channel surround sound. It also incorporates the EAX Advanced HD technology for truly immersive PC gaming experience and Dolby Digital EX compatibility, among others. Be it gaming or listening to

music, you can enjoy rich and realistic 6.1 surround sound. It utilises the breakthrough surround sound technology of Windows Media Audio 9, bringing in the next wave of home theatre-like experience to the PC. It has an incredible 106 dB Sound to Noise Ratio and the unique reputation of being the first and only THX certified add-in soundcard.



Amazon red-faced

Amazon's online store lets shoppers host recommendations for other buyers shopping for the same merchandise. Now, this technology, that tracks and links consumer perusal and purchases, has left Amazon red in the face due to its latest *faux pas*.

In early December 2002, Amazon users who were looking at a spiritual guide by well known Christian evangelist Pat Robertson were shocked to find a manual on sex rec-

ommended alongside. The incident clearly illustrates the pitfalls of online recommendations, which were hailed as excellent marketing tools. After receiving a complaint, Amazon promptly withdrew the recommendation. Unfortunately, this technology is easily manipulated by people who repeatedly shop, or just click through the links online. Analysts claim that this sort of incident will drive away existing and potential customers.

